

PUJITHA ATTULURI

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Data Engineer | Analytics Engineer | Data Scientist

SUMMARY

Data engineer with 4+ years of production experience building enterprise-scale ETL pipelines, BI platforms, and GenAI systems at Mu Sigma by cutting SLA processing time by 60% and driving \$48M in measurable business savings. Most recently engineered the Microsoft Fabric migration for Hearst Television's revenue reporting platform as a Business Intelligence Intern, while completing an M.S. in Data Science & Business Analytics at UNC Charlotte (GPA 4.0, Dec 2026). Azure-certified and fluent across the modern data stack : Databricks, Delta Lake, Power BI, and production RAG/LLM architectures with hands-on research experience in AI-generated media detection.

EDUCATION

University of North Carolina at Charlotte

M.S., Data Science & Business Analytics — GPA 4.0

Charlotte, NC

Aug 2025 – Dec 2026

SASTRA University

B.Tech, Information Technology — GPA 3.5

India

Jul 2016 – May 2020

TECHNICAL SKILLS

Cloud & Data Platform: Azure Data Factory, Azure Databricks, Microsoft Fabric, Azure Synapse, Snowflake, AWS (S3, EC2), Docker

Data Engineering: Apache Spark, PySpark, Delta Lake (ACID, SCD Type 2, versioning), Lambda Architecture, Databricks Autoloader, Kafka, Hadoop, ETL/ELT, batch & streaming pipelines

Databases: Advanced SQL, MySQL, Azure SQL DB, Redis, NoSQL

BI & Visualization: Power BI (DAX, semantic models, RLS), Power Automate, Plotly Dash, Streamlit

AI & GenAI: RAG architecture (LangChain, FAISS, pgvector), LLM evaluation & fine-tuning, prompt engineering, Scikit-learn, XGBoost

DevOps & Programming: Python, SQL, R, Git, Azure DevOps, CI/CD, SonarQube

Certifications: Microsoft Azure Data Engineer Associate, Snowflake Data Warehouse Essentials

WORK EXPERIENCE

Business Intelligence Intern, HTV IQ Platform — Hearst Television

Jun 2026 – Sep 2026

Charlotte, NC

- Contributed to the enterprise BI modernization migrating legacy Decentrix infrastructure to Microsoft Fabric, developed semantic data models with validated fact/dimension relationships and implemented Row-Level Security across Hearst Television stations.
- Authored SQL-based validation and discrepancy-reporting frameworks in Microsoft Fabric to reconcile migrated revenue data against the legacy platform across Stations, Advertisers, Agencies, and Account Executives, automating reconciliation across Dev, Test, UAT, and Production environments.
- Monitored Fabric pipeline refresh timing and optimized validation windows for near-real-time revenue reporting, collaborating in an Agile process to refine semantic models and dashboard logic.

Research Assistant, AI-Generated Music Detection — UNC Charlotte, School of Data Science

May 2026 – Jun 2026

Charlotte, NC

- Built an end-to-end Python pipeline to scrape Billboard chart data, extract audio features via the Gemini API (MFCCs, spectral centroid, chroma), and generate synthetic AI music through Suno and Udio APIs, constructing a balanced training dataset for a music-authenticity classifier.
- Designed schema validation and quality checkpoints across each pipeline stage, applying systematic prompt engineering across genres and production styles to reduce distribution bias for downstream CNN, wav2vec, and CLAP model training.

Teaching Assistant / Graduate Assistant — UNC Charlotte, School of Data Science

Jan 2026 – May 2026

Charlotte, NC

- Supported faculty in delivering data science coursework, mentoring students through Python, SQL, Power BI, and Streamlit projects, and guiding them through end-to-end analytics work with structured feedback on code quality and analytical rigor.
- Produced written instructional materials that made complex technical concepts accessible to students from diverse technical backgrounds.

Apprentice Leader — Mu Sigma Inc.

Jan 2024 — Jan 2025

Bangalore, India

- Spearheaded a feature team of data engineers and analysts, translating ambiguous project requirements into technical and business solutions and resolving cross-functional blockers.
- Led development of a supply-chain risk simulation web app on Azure Web App Service, enabling users to model supply disruptions and generating \$48M in year-to-date savings by mitigating production risk.
- Implemented Slowly Changing Dimension (Type 2) using Delta Lake with full ACID compliance, cutting computational costs by 60%.
- Developed a production GenAI RAG framework leveraging LLMs, reducing AI compute costs by 15% and shaping the product roadmap to align initiatives with organizational goals.

Decision Scientist — Mu Sigma Inc.

Jan 2021 — Jan 2024

Bangalore, India

- Redesigned the data storage schema for the Exceptions Planning project and migrated Parquet files to Delta format with multithreading and version history, cutting SLA processing time by 60%.
- Built high-frequency data models using Lambda Architecture and Databricks Autoloader, delivering near-real-time data to planners, reducing planning cycle times, and boosting inventory accuracy by 20%.
- Designed a scalable Power BI reporting layer processing 100M+ records per cycle, saving 6+ hours per week for each inbound scheduler across all sites.
- Migrated Databricks ETL workflows from interactive to job clusters, reducing cluster management costs by 15% and minimizing overload failures across products.

PROJECTS

Steam Game Success Predictor *Python, XGBoost, SHAP, Optuna, Groq API, Llama 3, RAG, Streamlit*

- Engineered a custom success metric from 40,000+ Steam games; built and compared four models (Random Forest, XGBoost, Optuna-tuned XGBoost, Stacking Ensemble) with SHAP analysis to quantify drivers of commercial success. Shipped a 3-tab Streamlit app with live inference, a RAG-powered strategy chatbot, and a market-segmentation dashboard. Team project.

Osteoporosis Care Gap Analysis *Python, PCA, KNN, Scikit-learn, SQL, Streamlit*

- Applied PCA-based clustering and KNN classification to a clinical dataset, identifying care-gap status as the strongest predictor of repeat-fracture risk (ROC-AUC 0.94) and quantifying up to \$1.3M in savings at 75% referral adoption.

Netflix Churn Prediction *Python, Scikit-learn, PySpark, Feature Engineering*

- Built an end-to-end churn classification pipeline with iterative feature engineering and failure-mode diagnosis, improving at-risk subscriber recall by 23% and achieving a 0.89 ROC-AUC. Team project.

Bowtie Risk Diagram *Python, JavaScript, ReactFlow, Streamlit*

- Refactored a 650-line monolithic codebase into a modular 17-file architecture; built and deployed an interactive rideshare safety risk-analysis app with ELK auto-layout and collapsible reasoning logic to Streamlit Cloud. Team project.